Non-small cell lung cancer

Adenocarcinoma · squamous · large cell · **TNM 9th edition (Jan 2025)** · **broad molecular testing at diagnosis**

Compiled from the Stanford Oncology Handbook 2025, NSCLC (Salter A, Roy M) · NCCN · primary trial reports verified against PubMed, September 2026.

WORKUP

- **Chest CT with contrast** (nodes are far easier to call with contrast)
- **Brain MRI with contrast for every stage except IA** — PET never substitutes for it
- **PET/CT** for distant disease (or CT C/A/P + bone scan if unavailable)
- Biopsy: EBUS, CT-guided transthoracic, metastatic site, or pleural cytology. **Non-decalcified cores preferred for molecular**
- **Pathologic mediastinal nodal evaluation** for non-metastatic disease, especially stage II–III. EBUS reaches 2R/2L, 4R/4L, 7, 10, 11; **mediastinoscopy cannot reach stations 5–6** (needs a Chamberlain procedure)
- **PD-L1 in every case.** Molecular sequencing in all adenocarcinoma; add for squamous if never/light smoker, small biopsy, or adenosquamous
- **Stage IIIB or IV: concurrent tissue + liquid molecular testing** (ctDNA ~7-day turnaround). **A negative liquid biopsy does not rule out a driver** — confirm on tissue

STAGING — TNM 9TH (JAN 2025)

	Definition
T1 / **T2**	≤3 cm / >3–5 cm, or visceral pleura, adjacent lobe, or main bronchus
T3 / **T4**	>5–7 cm or chest wall, phrenic nerve, same-lobe nodule / **>7 cm** or mediastinum, great vessels, carina, esophagus, spine, different ipsilateral lobe
N1 / **N2** / **N3**	Ipsilateral hilar (stations 10–14) / ipsilateral mediastinal or subcarinal — **N2a single station, N2b multiple** / contralateral or any supraclavicular
M1a / **M1b**	Pleural-pericardial effusion or nodules, or contralateral lung nodule / **single** extrathoracic met
M1c1 / **M1c2**	Multiple mets, **single organ system** / **multiple organ systems**

What changed from the 8th edition: N2 split into N2a/N2b and M1c split into M1c1/M1c2. **T descriptors are unchanged.**

TREATMENT PATHWAY

Resectable — stage I to III

Decide resectability FIRST
before any neoadjuvant therapy, at tumor board
systemic therapy if **stage I >4 cm, stage II, stage III**

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Perioperative immunotherapy
CheckMate 816 (neoadjuvant only) · KEYNOTE-671
AEGEAN · CheckMate 77T (perioperative)
omit IO if EGFR / ALK / RET / ROS1 altered

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Adjuvant (no neoadjuvant given)
retest the specimen: EGFR, ALK, PD-L1
platinum doublet ×4, then IO if PD-L1 ≥1%
EGFR → osimertinib ×36 mo (ADAURA)
ALK → alectinib ×24 mo (ALINA)

Unresectable stage III

Concurrent chemoradiation
non-squamous: carboplatin AUC 5 + pemetrexed
squamous: cisplatin/etoposide or weekly carbo/paclitaxel

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Consolidation
durvalumab within 6 weeks, ×12 months (PACIFIC)
EGFR-mutant: osimertinib instead (LAURA) —
PFS **39.1 vs 5.6 months**, HR 0.16

Metastatic

Wait for molecular results
do not start systemic therapy until they return
IO before excluding a driver worsens outcomes
if severely symptomatic: 1–2 cycles platinum, then tailor

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Then treat by result
driver positive → targeted therapy, whatever the PD-L1
PD-L1 ≥50%, no driver → single-agent IO
any PD-L1 → platinum doublet + pembrolizumab

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Oligometastatic (≤3–5 sites)
local consolidative therapy (SBRT or surgery) to all
sites, plus systemic therapy — improves PFS and OS

SCREENING

- **USPSTF 2021 (Grade B):** annual **low-dose CT**, age **50–80**, **≥20 pack-years**, currently smoking **or quit within 15 years**
- Stop at **15 years since cessation**, or when comorbidity would preclude curative surgery
- **NLST:** 20% reduction in lung-cancer mortality vs chest radiography · **NELSON:** 24% in men
- **Smoking cessation remains the highest-yield intervention** — duration of smoking matters more than pack count

HISTOLOGY

Type	Features
Adenocarcinoma	~50%; peripheral, non/light smokers. **TTF-1, napsin A, CK7** positive. **Highest yield for actionable drivers.** Mucinous (TTF-1–) is KRAS-enriched and EGFR-poor
Squamous	~25%; central, cavitating, heavy smokers. **CK5/6, p40** / **p63**; keratin pearls and intercellular bridges. **Pemetrexed and bevacizumab are both contraindicated**
Large cell	Diagnosis of exclusion — treat as non-squamous
Adenosquamous	Both components, each ≥10% of the tumor

EARLY STAGE — SURGERY AND RADIATION

- **Lobectomy + mediastinal nodal sampling or dissection** is the standard; VATS preferred over thoracotomy
- **Sublobar resection is acceptable for peripheral cT1aN0 ≤2 cm**, after intraoperative nodal confirmation. **CALGB 140503:** non-inferior DFS (HR 1.01). **JCOG0802:** segmentectomy 5-yr OS 94.3% vs 91.1% (from fewer non-cancer deaths) **but double the local relapse** (10.5% vs 5.4%)
- **SBRT** for medically inoperable stage I. **Avoid central SBRT within 2 cm of the proximal bronchial tree**
- Preoperative assessment: **predicted postoperative FEV1 and DLCO**; <30% predicted flags high surgical risk

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DRIVER → THERAPY

Alteration	First line and beyond
EGFR ex19del / L858R *~15% US adeno*	**Osimertinib 80 mg daily** (FLAURA). **Add carbo/pemetrexed ×4** for higher-risk disease — CNS involvement, L858R, TP53 co-mutation (FLAURA2: mPFS **25.5 vs 16.7 mo**; OS 47.5 vs 37.6). **Or amivantamab + lazertinib** (MARIPOSA: mPFS **23.7 vs 16.6 mo**, HR 0.70; **OS HR 0.75**) — but grade ≥3 AEs **80% vs 52%**, and it needs **VTE prophylaxis for 4 months**
EGFR exon 20 ins *~10% of EGFR*	**Resistant to osimertinib and the classic TKIs.** 1L **amivantamab + carboplatin/pemetrexed** (PAPILLON: mPFS **11.4 vs 6.7 mo**, HR 0.40). Post-platinum: **sunvozertinib** (accelerated, **2L+ only**). **Mobocertinib withdrawn**
ALK *~3–7%*	**Lorlatinib 100 mg daily** (CROWN: 5-yr PFS **60% vs 8%** vs crizotinib, median not reached — the longest single-agent PFS reported in any solid tumor). Watch **CNS effects** (mood, cognition, seizures) and **hyperlipidemia**. Alternatives: alectinib, brigatinib, ensartinib
ROS1 *~1–2%*	**Four co-preferred 1L agents:** crizotinib, entrectinib, repotrectinib, taletrectinib. **Prefer entrectinib / repotrectinib / taletrectinib if brain metastases** — crizotinib has no CNS activity. Repotrectinib TKI-naive ORR 79%, mPFS 35.7 mo. **Alectinib has no ROS1 activity** despite its ALK potency
RET *~1–2%*	**Selpercatinib** — the only randomized 1L evidence (LIBRETTO-431: mPFS **24.8 vs 11.2 mo** vs chemo ± pembrolizumab, HR 0.47; intracranial ORR 82%). Pralsetinib is an alternative with more pneumonitis. Watch hypertension and transaminitis
MET exon 14 *~3–4%, older never-smokers*	**Capmatinib** (1L ORR 68%) or **tepotinib** (1L ORR ~57%) — single-arm, not comparable. Tepotinib: peripheral edema
BRAF V600E *~2%, smokers*	**Encorafenib + binimetinib** (PHAROS, treatment-naive ORR 75%) or **dabrafenib + trametinib** (ORR 64%, mPFS 10.9 mo). **Pyrexia is the classic combination toxicity** — hold and dose-modify
KRAS G12C *~13% of adeno*	**Not a first-line target** — 1L is chemo-IO. **2L after platinum + anti-PD-(L)1: adagrasib** (KRYSTAL-12: mPFS 5.5 vs 3.8 mo, HR 0.58) or **sotorasib** (CodeBreaK 200: 5.6 vs 4.5 mo, no OS gain; **accelerated approval only**)
HER2 (ERBB2) mutant *~2–3%, non-smoker women*	Mostly exon 20 insertions; **high rate of brain metastases at presentation**. **Zongertinib** — oral, and as of **Feb 2026 it carries a 1L indication** (treatment-naive ORR 76%). **T-DXd 5.4 mg/kg q3wk** (DESTINY-Lung02: ORR 49%) — **ILD in 13–15%, including fatal cases**; monitor with serial imaging
NTRK fusion *<1%*	Larotrectinib, entrectinib or repotrectinib — all tumor-agnostic approvals
NRG1 fusion	**Zenocutuzumab** — accelerated approval, **after progression on prior systemic therapy**, for NSCLC and pancreatic adenocarcinoma only

PROGRESSION ON OSIMERTINIB

- **Rebiopsy** — resistance is heterogeneous, with no single dominant mechanism
- **On-target:** EGFR **C797S**. **Off-target:** MET amplification, HER2 amplification, acquired fusions
- **Histologic transformation is the one you cannot afford to miss: SCLC transformation** (with RB1 and TP53 loss) or squamous transformation. **Treat SCLC transformation as SCLC** — platinum/etoposide
- **Standard next step is platinum-doublet chemotherapy**
- **Amivantamab + chemotherapy** (MARIPOSA-2: mPFS 6.3 vs 4.2 mo, HR 0.48). **Approved regimen excludes lazertinib** (hematologic toxicity)
- **Datopotamab deruxtecan** — TROP2 ADC, accelerated approval, **EGFR-mutant only**

DRIVER-NEGATIVE METASTATIC

- **PD-L1 ≥50%: pembrolizumab** monotherapy (KEYNOTE-024: OS 30 vs 14 months); atezolizumab or cemiplimab are alternatives. **Use chemo-IO instead if the burden is high or the patient is symptomatic**
- **PD-L1 1–49% or <1%:** chemo-IO doublet. **Non-squamous** → pembrolizumab + carboplatin + pemetrexed (KEYNOTE-189). **Squamous** → pembrolizumab + carboplatin + paclitaxel (KEYNOTE-407)
- **Dual IO** — ipilimumab/nivolumab (CheckMate 227, 9LA) or durvalumab/tremelimumab (POSEIDON): chemo-sparing but more toxic, **most attractive at PD-L1 <1% and in squamous**
- **Second line: docetaxel 35 mg/m² days 1+8** — split dosing is better tolerated than 75 q3wk — **± ramucirumab**

ADJUVANT THERAPY

- **Cisplatin doublet ×4** for stage II–IIIA (LACE: ~5% absolute 5-yr OS gain). **Stage IB only if ≥4 cm or high-risk features**; **no benefit and possible harm in stage IA**
- **EGFR → osimertinib ×36 months** (ADAURA: DFS HR 0.17, OS HR 0.49). **ALK → alectinib ×24 months** (ALINA: DFS HR 0.24). **Omit IO in both**
- **Atezolizumab** (IMpower010) is approved for **stage II–IIIA with PD-L1 ≥1%**, EGFR/ALK wild-type — OS NS overall; benefit concentrated at PD-L1 ≥50%
- **Pembrolizumab** (KEYNOTE-091) after adjuvant chemotherapy, **no PD-L1 restriction**: DFS HR 0.76 (PD-L1 ≥50% subgroup NS)

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PERIOPERATIVE AND ADJUVANT DOSING

Regimen	Dosing
CheckMate 816 *neoadjuvant only*	Cisplatin **75 mg/m²** + pemetrexed **500 mg/m²** (or gemcitabine **1250** d1+8) + **nivolumab 360 mg** on day 1 — **q21d ×3**, then surgery
KEYNOTE-671 *perioperative*	Cisplatin 75 + pemetrexed 500 (or gemcitabine 1000 d1+8) + **pembrolizumab 200 mg** q21d ×4 → surgery → within 12 weeks, **pembrolizumab 200 mg q21d ×13**
AEGEAN *perioperative*	Cisplatin 75 + pemetrexed 500 (or gemcitabine 1250 d1+8) + **durvalumab 1500 mg** q21d ×4 → surgery → **durvalumab 1500 mg q28d ×12**
CheckMate 77T *perioperative*	Cisplatin 75 + pemetrexed 500 (or docetaxel 75) + **nivolumab 360 mg** q21d ×4 → surgery → within 90 days, **nivolumab 360 mg q28d ×1 year**
Adjuvant chemo	Cisplatin **75 mg/m²** + pemetrexed 500 (non-squamous), gemcitabine 1250 d1+8, or docetaxel 75 — **q21d ×4**. Growth factor support if using docetaxel
Adjuvant IO	**Atezolizumab 1680 mg q4wk** or **pembrolizumab 200 mg q3wk**, after adjuvant chemotherapy

CHEMORADIATION AND METASTATIC DOSING

Regimen	Dosing
ChemoRT non-squamous	Carboplatin **AUC 5** + pemetrexed **500 mg/m²** q21d ×4, with daily RT
ChemoRT squamous	Cisplatin **50 mg/m² days 1, 8, 29, 36** + etoposide **50 mg/m² days 1–5 and 29–33**; or carboplatin **AUC 2** + paclitaxel **45 mg/m²** weekly — with daily RT
Consolidation durvalumab	**10 mg/kg q14d ×12 months** (or 1500 mg q4wk). **Start within 6 weeks** of finishing chemoRT
Platinum doublet + IO	Carboplatin **AUC 5–6** + pemetrexed 500 (non-squamous) or paclitaxel 200 (squamous) + **pembrolizumab 200 mg** q21d ×4 → pemetrexed + pembrolizumab maintenance
Single-agent IO	Pembrolizumab **200 mg q3wk** (or 400 q6wk) · atezolizumab **1200 mg q3wk** · cemiplimab **350 mg q3wk**
Osimertinib + chemo *FLAURA2*	Carboplatin **AUC 6** + pemetrexed **500** q21d ×4, with **osimertinib 80 mg daily continuously**, then pemetrexed maintenance
Amivantamab + lazertinib	Amivantamab weight-based (**<80 kg 1050 mg**, **≥80 kg 1400 mg**) — split over days 1–2 in week 1, weekly through week 5, then **q2wk from week 7**; + **lazertinib 240 mg PO daily**. **VTE prophylaxis ×4 months**
Docetaxel, 2L	**35 mg/m² days 1+8 q21d** — preferred over 75 q21d; reduce to 30 or 25 if heavily pretreated. **± ramucirumab 10 mg/kg**
Pemetrexed maintenance	**500 mg/m² q21d** — non-squamous only

MONITORING AND CAUTIONS

- **Baseline HBsAg + anti-HBc before any immunosuppressive therapy**; consider quantitative TB testing by risk
- **Pemetrexed:** folic acid from ≥1 week before until 3 weeks after · B12 1000 mcg q9wk · dexamethasone 4 mg BID the day before and after, for rash. **Never in squamous**
- **Bevacizumab:** contraindicated in **squamous** histology, significant hemoptysis, and large cavitary tumors. Urine protein every other cycle; treat hypertension
- **Cisplatin:** avoid with hearing loss, CrCl <30, severe neuropathy, or poor performance status. Extensive hydration and aprepitant-containing antiemetics
- **Osimertinib:** echo at baseline and q3 months ×6 months; ECG at baseline, 2 weeks, 6 weeks, and with any QT-prolonging drug
- **Alectinib:** CK at baseline, 2 weeks, then q3 months · LFTs q2 weeks ×3 months. **Lorlatinib:** CNS effects and hyperlipidemia — median onset **15 days**; rosuvastatin first-line
- **Immunotherapy:** TSH and free T4 at baseline then q4–6 weeks. Prednisone **1–1.5 mg/kg** for irAEs with a slow taper
- **Do not start a targeted agent — osimertinib and crizotinib above all — within 3 months of immunotherapy**; the toxicity is additive. IO is also less effective in EGFR/ALK-driven disease

KEY TRIALS

CheckMate 816	Neoadjuvant platinum + nivolumab ×3 — pCR 24% vs 2%, EFS HR 0.63, OS HR 0.62
KEYNOTE-671	Perioperative pembrolizumab — EFS HR 0.58; the first perioperative IO trial with a confirmed OS benefit
AEGEAN / 77T	Perioperative durvalumab (EFS HR 0.68) / nivolumab (EFS HR 0.58)
ADAURA / ALINA	Adjuvant osimertinib ×36 mo (DFS HR 0.17, OS HR 0.49) / alectinib ×24 mo (DFS HR 0.24)
PACIFIC	Consolidation durvalumab after chemoRT — 5-yr OS **43% vs 33%**
LAURA	Consolidation osimertinib, EGFR-mutant stage III — PFS **39.1 vs 5.6 mo**, HR 0.16
FLAURA / FLAURA2	Osimertinib 1L; adding chemo gives mPFS 25.5 vs 16.7 mo and OS 47.5 vs 37.6
MARIPOSA	Amivantamab + lazertinib vs osimertinib — mPFS 23.7 vs 16.6 mo, **OS HR 0.75**, at a real toxicity cost
CROWN	Lorlatinib 1L in ALK — **5-yr PFS 60% vs 8%**
PAPILLON	Amivantamab + chemo 1L for EGFR exon 20 — mPFS 11.4 vs 6.7 mo
LIBRETTO-431	Selpercatinib vs chemo ± pembrolizumab in RET — mPFS 24.8 vs 11.2 mo
CALGB 140503	Sublobar resection non-inferior for peripheral cT1aN0 ≤2 cm — DFS HR 1.01